POWER BASE / REMOVABLE MCU SUB-BOARD / REV D REVIEW

Common-ground plug-in controller. Power base retains one three-phase bridge, brake, analogue sensing and all hardware protection.
12-60 V bus / DC input <=10 A / phase 10 Arms design target / separate 12-24 V auxiliary.

Auxiliary input

eFuse / separate 12-24 V auxiliary

文件: aux_input.kicad_sch

Driver supply

12 V buck-boost / driver power good

文件: supply_12v.kicad_sch

Logic and sensors

5 V / local 3.3 V / base supervisor

文件: supply_logic.kicad_sch

Bus and brake

12-60 V bus / one external-resistor brake

文件: bus_brake.kicad_sch

U half bridge

U driver / TDSON8 half bridge / shunt

文件: bridge_u.kicad_sch

V half bridge

V driver / TDSON8 half bridge / shunt

文件: bridge_v.kicad_sch

W half bridge

W driver / TDSON8 half bridge / shunt

文件: bridge_w.kicad_sch

Current sensing

3 x INA241 / 1.65 V reference

文件: current_sense.kicad_sch

Voltage and NTC

DC bus / three phases / NTC

文件: voltage_sense.kicad_sch

Analog protection

Current windows / bus OV / driver UV

文件: comparators.kicad_sch

Hardware latch

Health chain / latch / reset / FAULT_N

文件: fault_logic.kicad_sch

Hardware PWM gates

Six PWM AND gates / local pull-downs

文件: control_interface.kicad_sch

Base interlock

CTRL_READY / ENABLE_QUALIFIED / BASE_READY

文件: base_interlock.kicad_sch

Base LEDs

D3 power lamp only

文件: base_leds.kicad_sch

POWER-CORE HEADERS

2 x 20 pins / 5 V / common GND / 3.3 V signals

文件: power_interface.kicad_sch

MCU / SWD / TTL / CAN / encoder / SPI live on the removable controller board. No shared NRST net.
BASE_LOGIC_OK is local base-supply reset permission, not the controller NRST.
CTRL_READY LOW or missing controller blocks ENABLE; BASE_READY separately reports local logic plus driver supply.
This schematic supersedes the archived integrated Rev C design; the old PCB is not valid for Rev D.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB

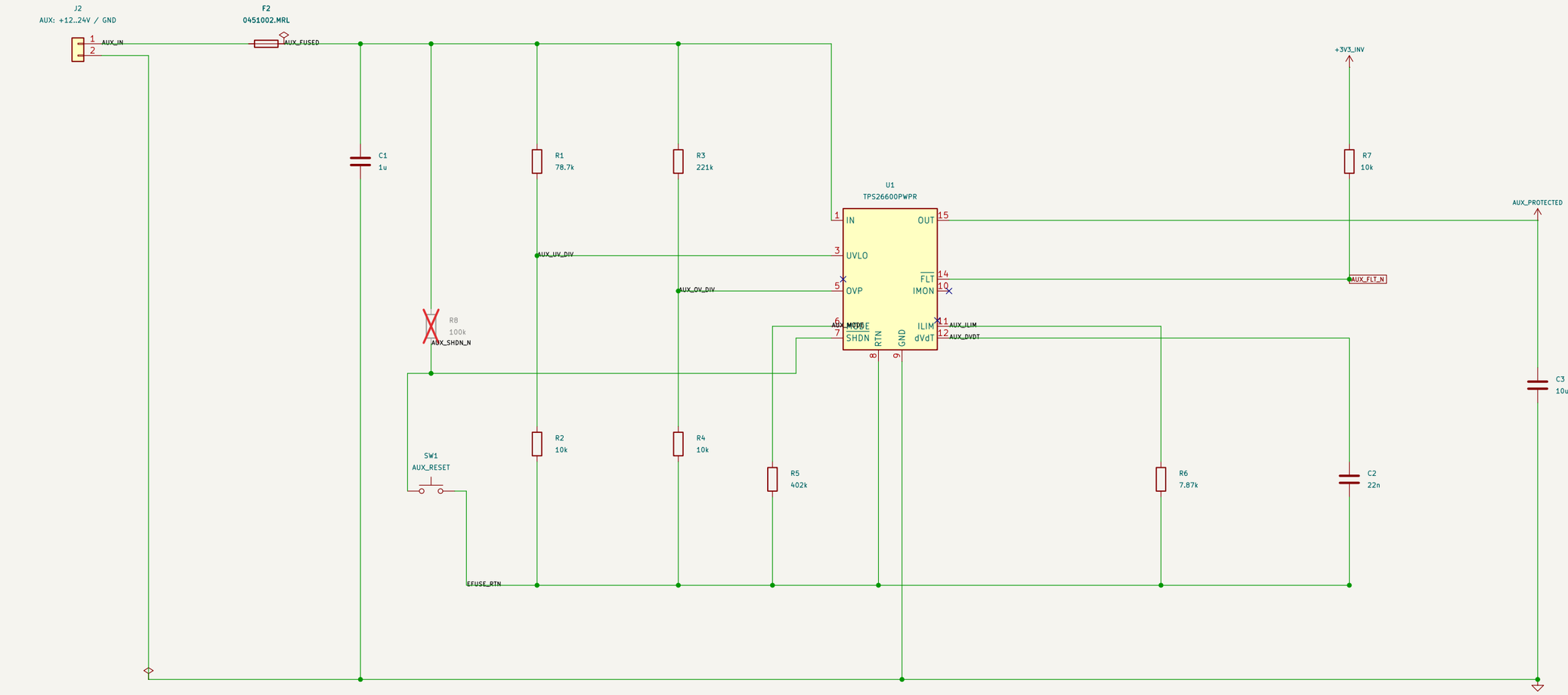
power_system

Sheet: /
File: inverter.kicad_sch

Title: Inverter power base - removable STM32G431 controller

Size: A3	Date: 2026-11-06	**Rev: D-split-draft**
KiCad E.D.A. 10.0.7		Id: 1/16

AUXILIARY INPUT / eFUSE / 12–24 V



RTN / exposed pad is the eFuse reference island. It is NOT the board GND pin.
MODE=402k selects latch-off. AUX_RESET resets the auxiliary path; no MCU startup dependency.
UV / OV / current-limit values remain unchanged; dynamics and surge coordination still require review.

INVERTER | non-isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT – NOT RELEASED FOR PCB

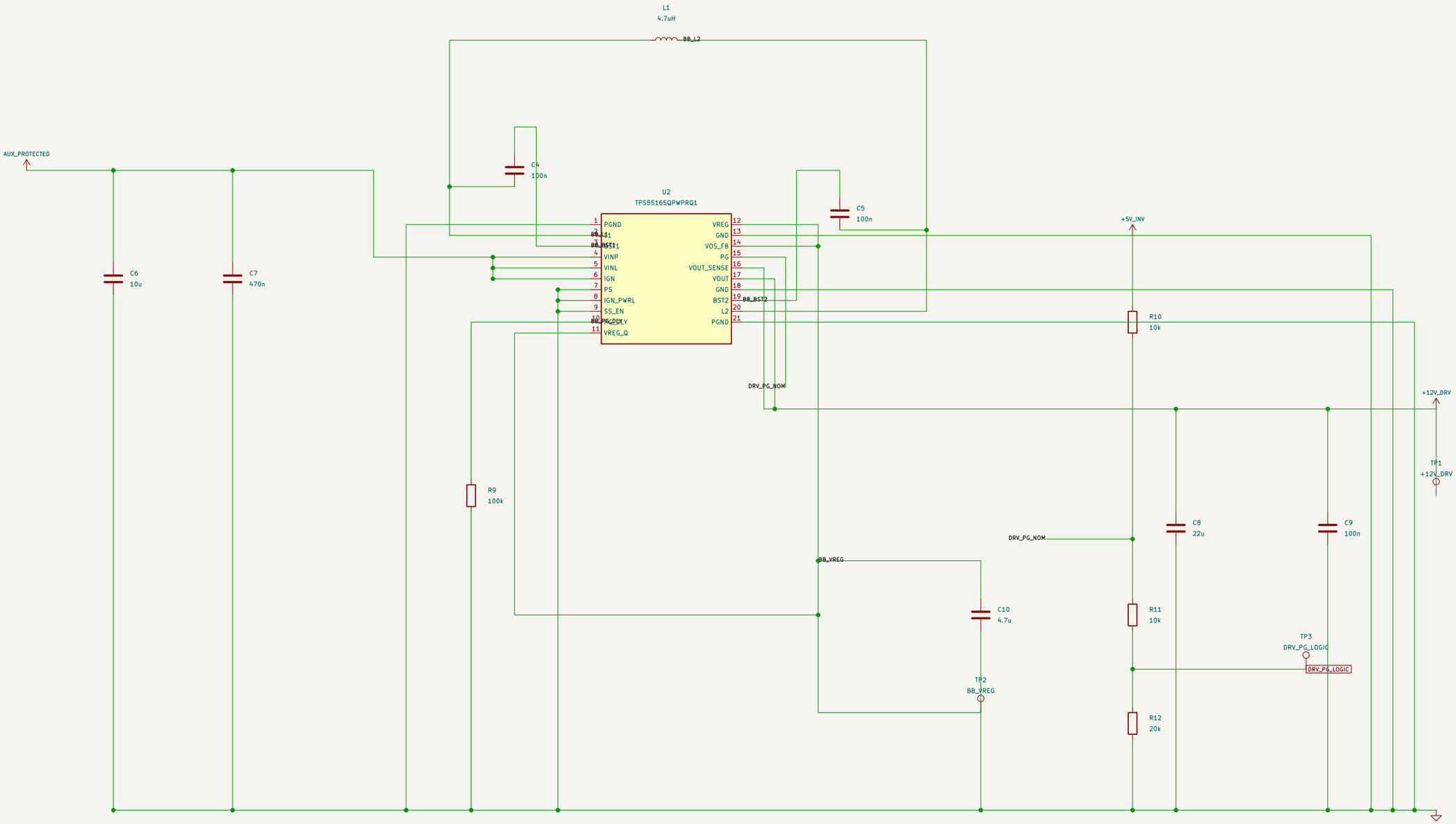
power_system

Sheet: /Auxiliary input/
File: aux_input.kicad_sch

Title: Inverter – 12–24 V auxiliary input protection

Size: A3	Date: 2026-11-06	**Rev: D-split-draft**
KiCad E.D.A. 10.0.7		Id: 2/16

12 V DRIVER RAIL / TPS55165 BUCK-BOOST



VREG_Q and VOS_FB return to VREG; VREG is not a load rail. VOUT_SENSE is tied to the 12 V output.
Input/output/bootstrap wiring is shown directly. PG high is about 2.5 V with the fitted pullup/divider.
Inductor model, startup, worst-case gate voltage and thermal capability remain review items.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB

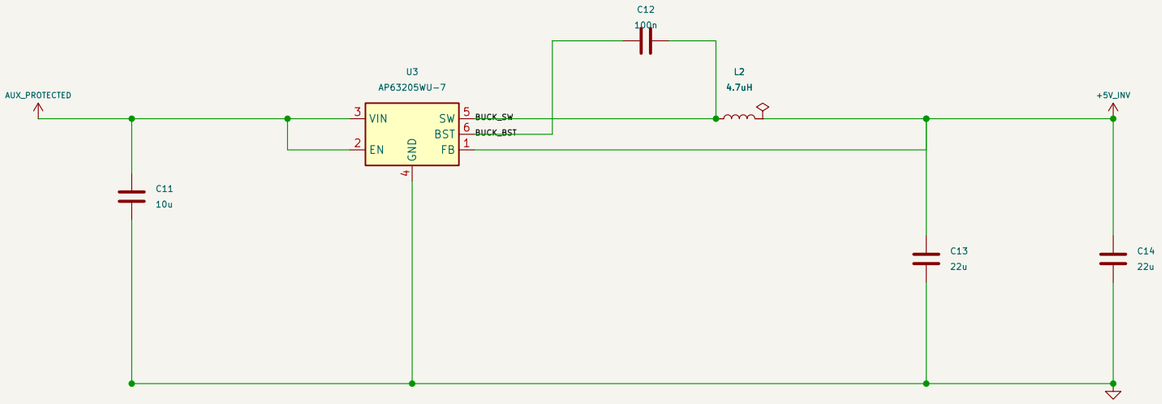
power_system

Sheet: /Driver supply/
File: supply_12v.kicad_sch

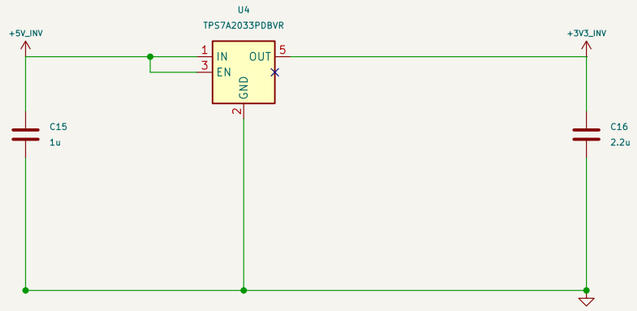
Title: Inverter - regulated 12 V driver supply

Size: A3	Date: 2026-11-06	**Rev: D-split-draft**
KiCad E.D.A. 10.0.7		Id: 3/16

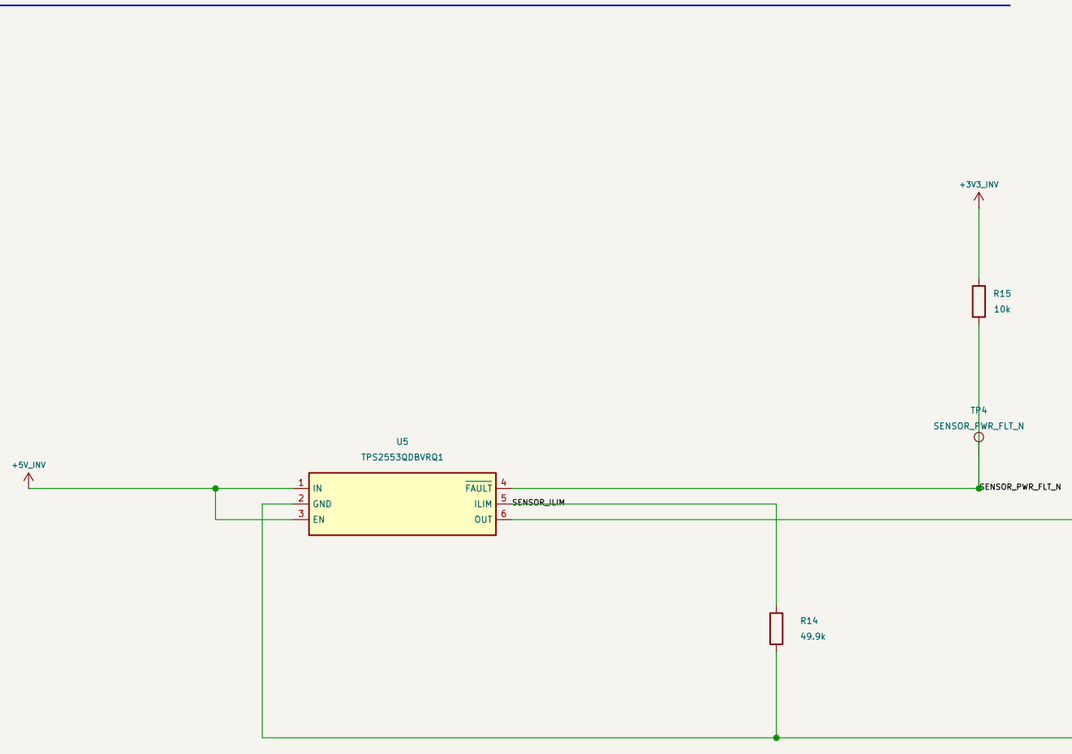
5 V BUCK / AP63205



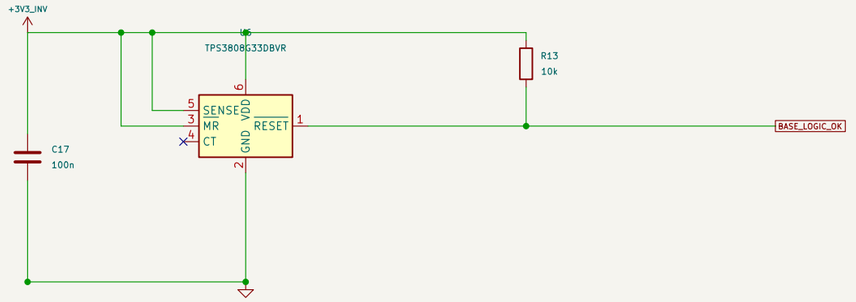
3.3 V BASE LOGIC / ANALOG SUPPLY



PROTECTED OPTIONAL SENSOR POWER / TOTAL TARGET 0.5 A



3.3 V SUPERVISOR / LOCAL BASE PERMISSION



J7 / external power links default DNP. The branch current target is total, not per connector.
U4 supplies only base analog and safety logic. BASE_LOGIC_OK never connects to controller NRST.
The removable MCU board regulates its own 3.3 V; local current budget and temperature require verification.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB

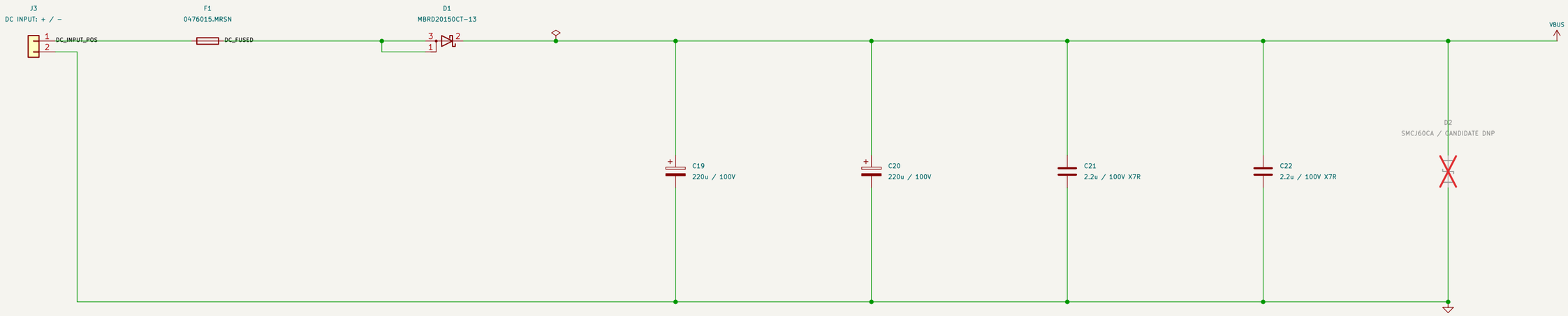
power_system

Sheet: /Logic and sensors/
File: supply_logic.kicad_sch

Title: Inverter - 5 V, 3.3 V and sensor power

Size: A3	Date: 2026-11-06	**Rev: D-split-draft**
KiCad E.D.A. 10.0.7		Id: 4/16

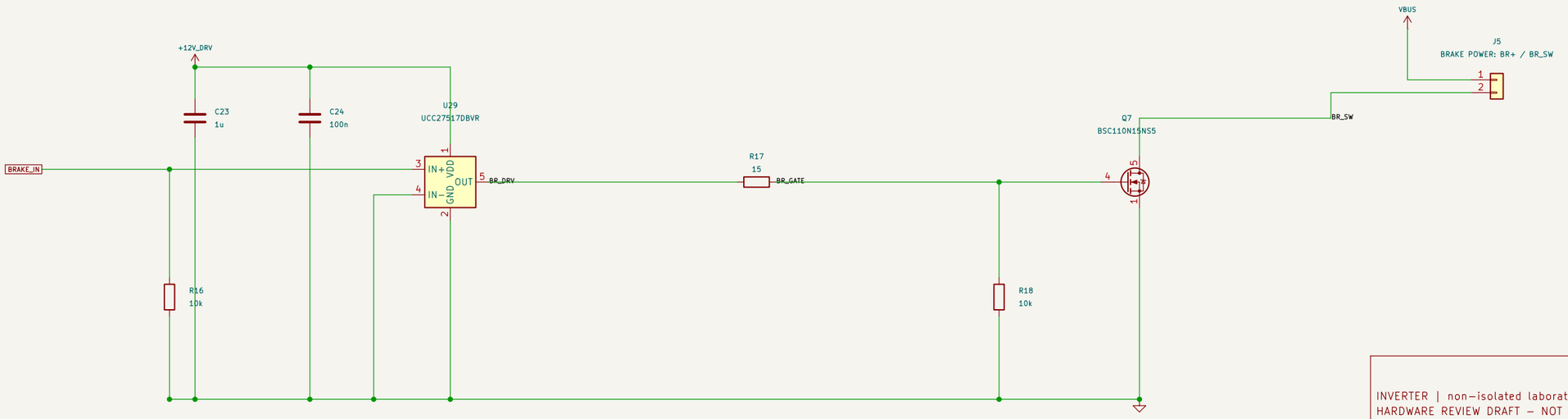
MAIN BUS INPUT / BULK CAPACITORS / RAW PWM OUTPUTS



D2 is DNP. The input diode / TVS / fuse are still procurement and energy/thermal review items.
No main-bus precharge or dedicated bleeder. U/V/W are raw bridge PWM, not LC-filtered sine outputs.



ONE BRAKE BRANCH / EXTERNAL RESISTOR / CONTROLLER HEADER COMMAND



J5: BR+ = VBUS, BR_SW = MOS drain. External resistor goes across these pins; unused port stays OPEN.
Brake is independent of the bridge ENABLE/fault gates. The base-side input pull-down defaults a missing controller command LOW.

INVERTER | non-isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT – NOT RELEASED FOR PCB

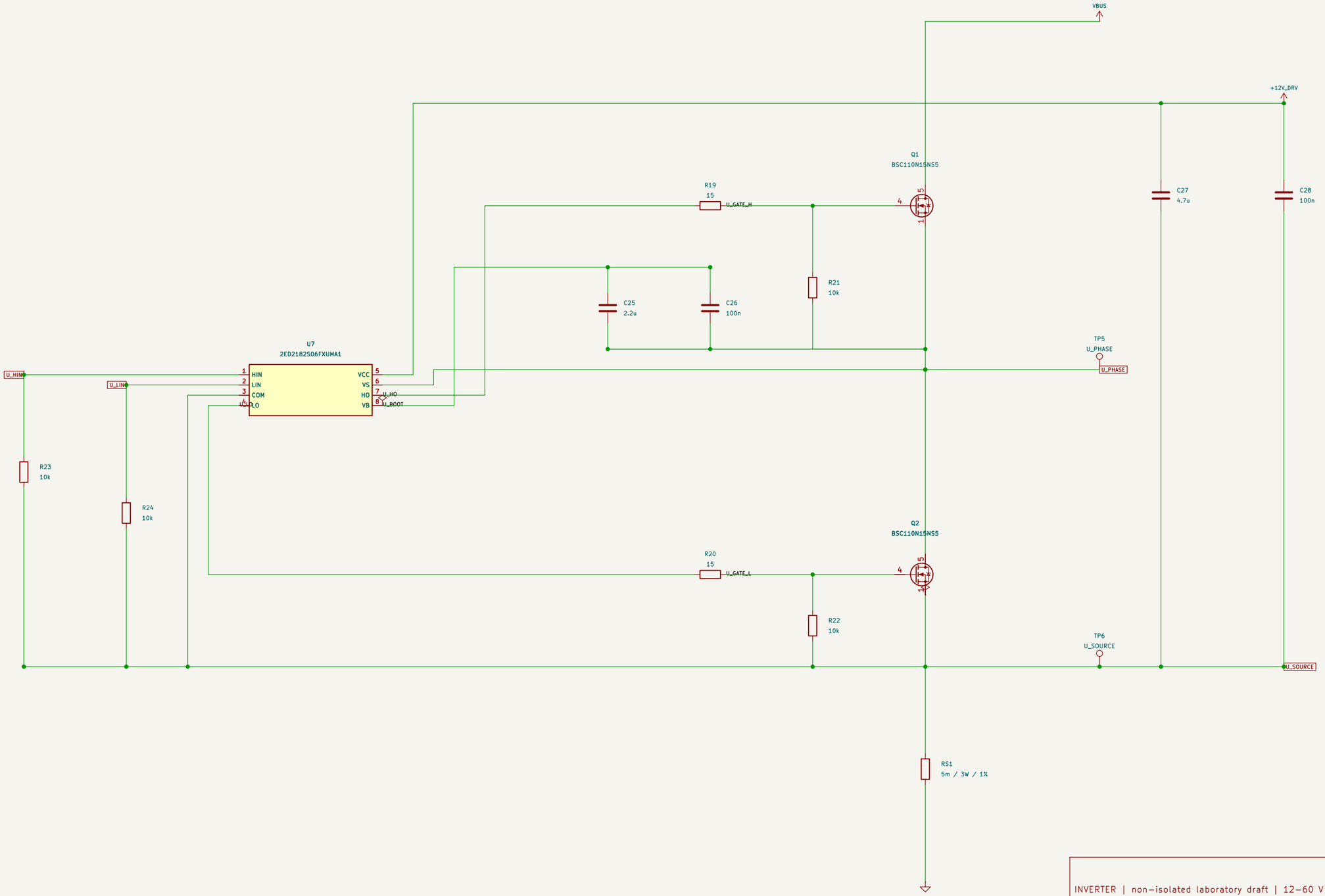
power_system

Sheet: /Bus and brake/
File: bus_brake.kicad_sch

Title: Inverter – DC bus input, bulk capacitors and brake

Size: A3	Date: 2026-11-06	**Rev: D-split-draft**
KiCad E.D.A. 10.0.7		Id: 5/16

U HALF BRIDGE / VISIBLE GATE, BOOTSTRAP AND SHUNT RETURNS



COM, low-gate pulldown, local VCC capacitors and sense high all return to SOURCE above the shunt.
High-gate pulldown and bootstrap capacitors return to PHASE. Common GND/DC- exists only below RS.
Gate R / bootstrap values remain initial values; deadtime, minimum recharge time and waveforms require validation.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB

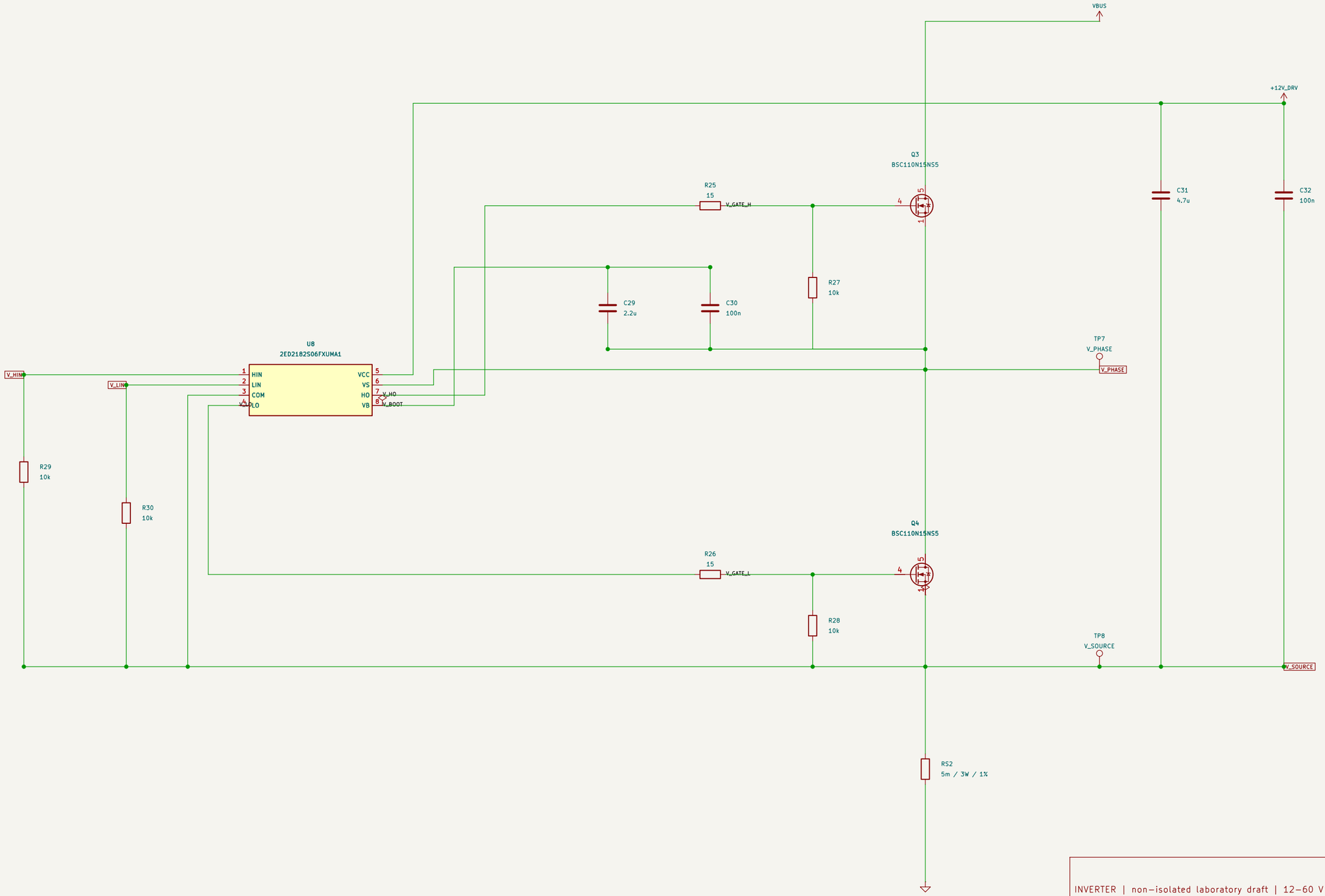
power_system

Sheet: /U half bridge/
File: bridge_u.kicad_sch

Title: Inverter - U half bridge and low-side shunt

Size: A3	Date: 2026-11-06	**Rev: D-split-draft**
KiCad E.D.A. 10.0.7		Id: 6/16

V HALF BRIDGE / VISIBLE GATE, BOOTSTRAP AND SHUNT RETURNS



COM, low-gate pulldown, local VCC capacitors and sense high all return to SOURCE above the shunt.
High-gate pulldown and bootstrap capacitors return to PHASE. Common GND/DC- exists only below RS.
Gate R / bootstrap values remain initial values; deadtime, minimum recharge time and waveforms require validation.

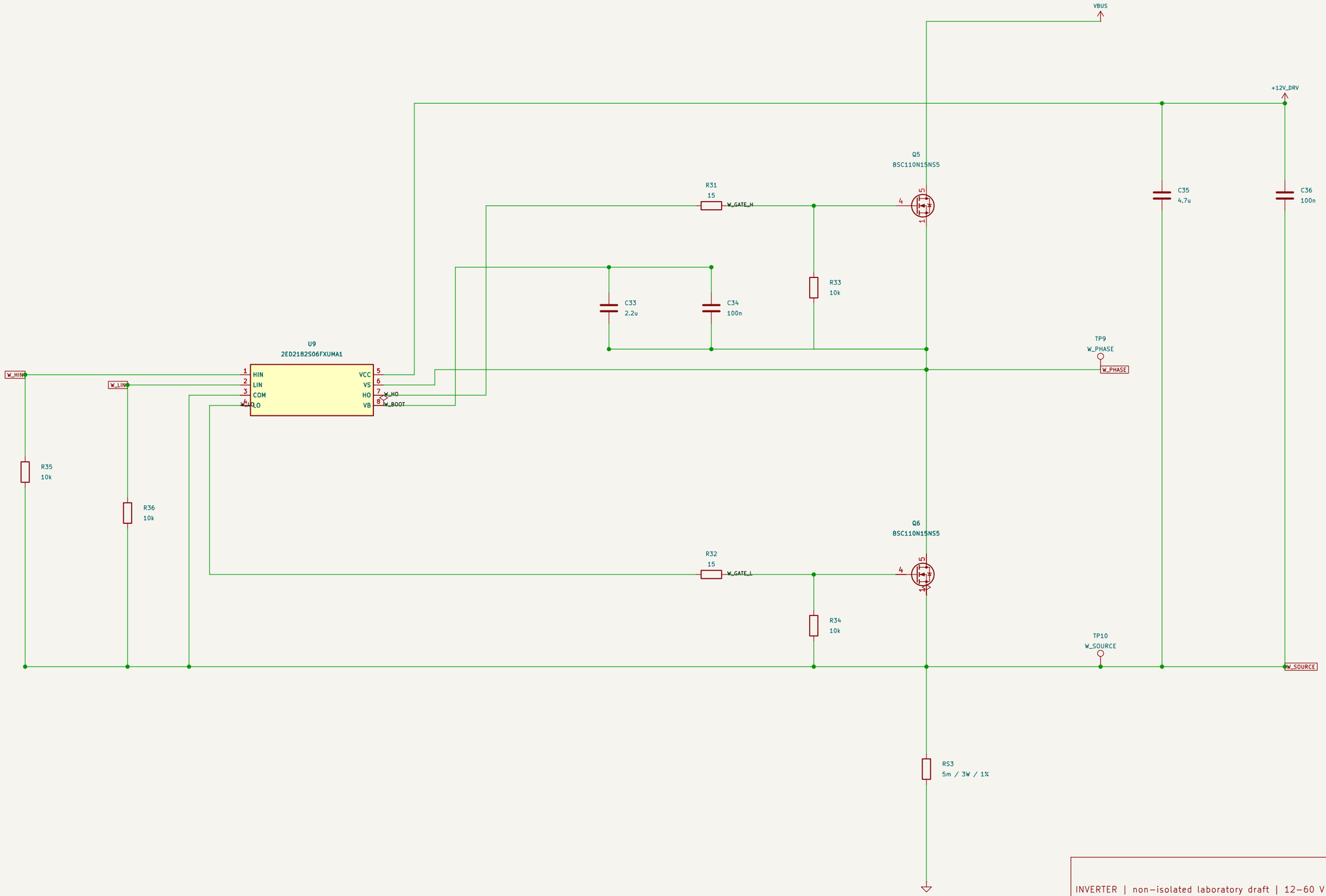
INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB
power_system

Sheet: /V half bridge/
File: bridge_v.kicad_sch

Title: Inverter - V half bridge and low-side shunt

Size: A3	Date: 2026-11-06	**Rev: D-split-draft**
KiCad E.D.A. 10.0.7		Id: 7/16

W HALF BRIDGE / VISIBLE GATE, BOOTSTRAP AND SHUNT RETURNS



COM, low-gate pulldown, local VCC capacitors and sense high all return to SOURCE above the shunt.
High-gate pulldown and bootstrap capacitors return to PHASE. Common GND/DC- exists only below RS.
Gate R / bootstrap values remain initial values; deadtime, minimum recharge time and waveforms require validation.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB

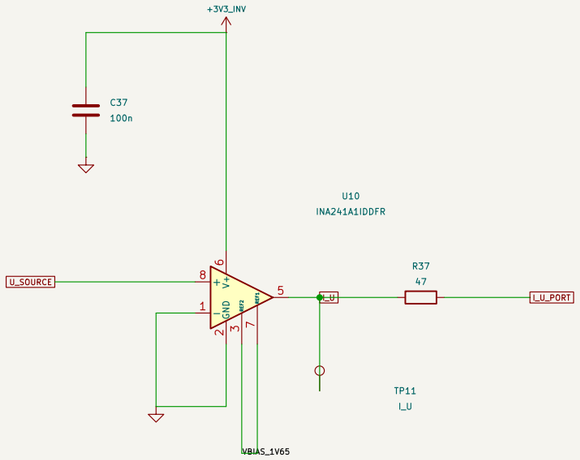
power_system

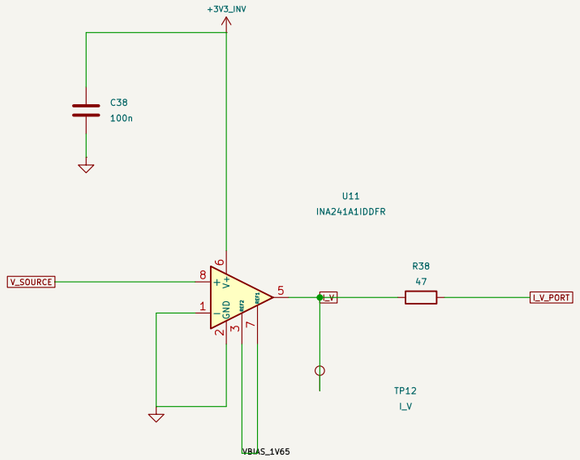
Sheet: /W half bridge/
File: bridge_w.kicad_sch

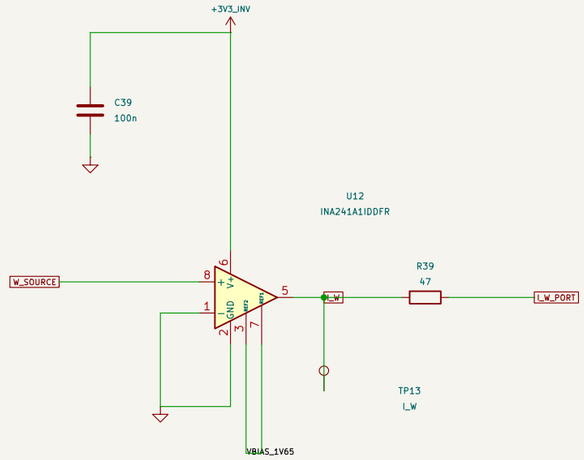
Title: Inverter - W half bridge and low-side shunt

Size: A3	Date: 2026-11-06	**Rev: D-split-draft**
KiCad E.D.A. 10.0.7		Id: 8/16

THREE LOW-SIDE KELVIN CURRENT CHANNELS

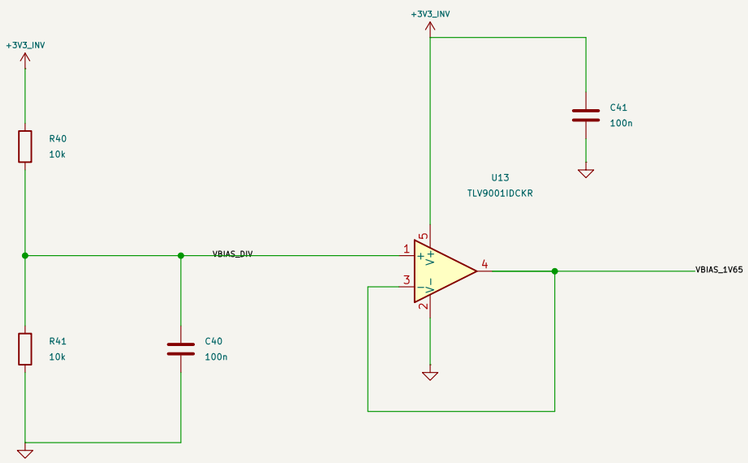


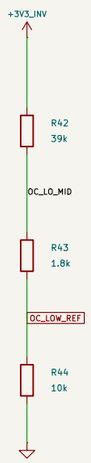


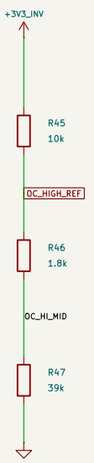


Each channel uses the buffered VBIAS_1V65 generated below.

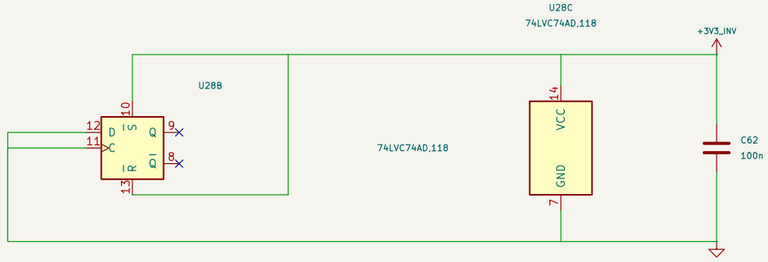
BUFFERED BIAS AND CURRENT WINDOW REFERENCES







SPARE U28B / FIXED INPUTS



I = 1.65 V + 50 mV/A; nominal current trips +/−20 A. Low-side shunts have switching-state blind intervals.
References track the local 3.3 V rail; phase-current sign and sampling windows require bench validation.

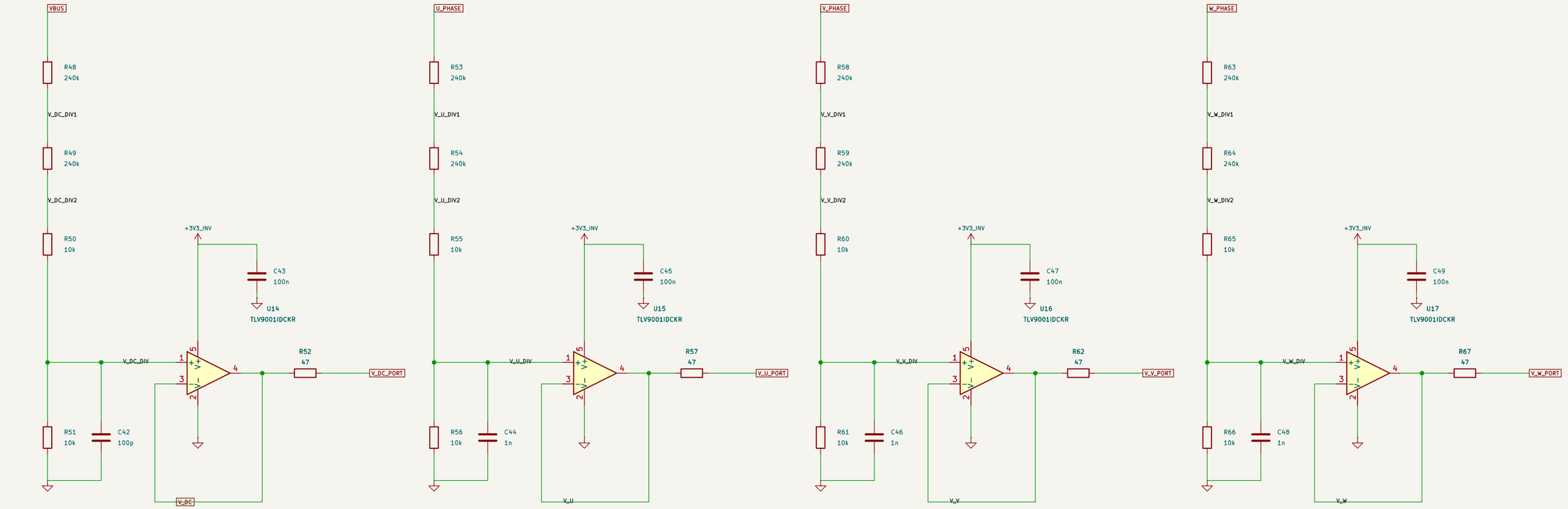
INVERTER | non-isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system

Sheet: /Current sensing/
File: current_sense.kicad_sch

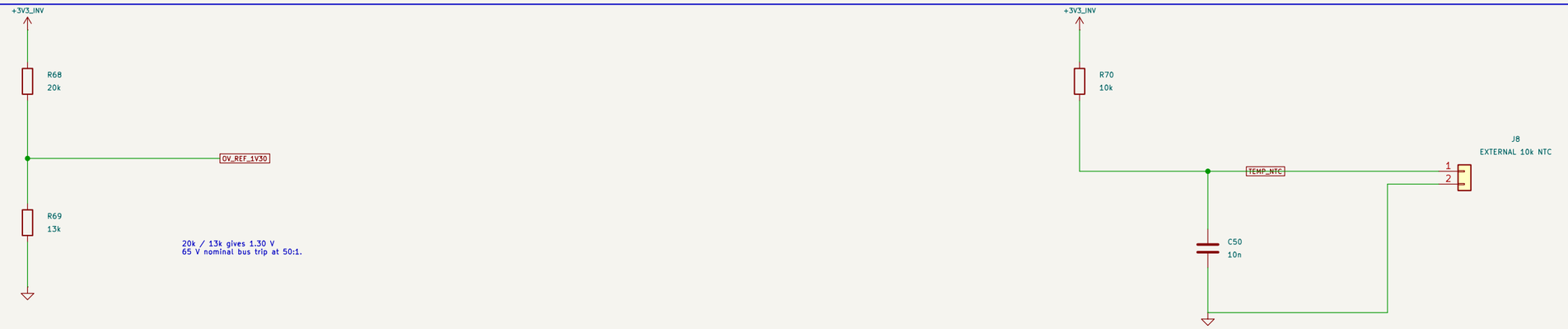
Title: Inverter — three current channels and references

Size: A3	Date: 2026-11-06	**Rev: D-split-draft**
KiCad E.D.A. 10.0.7		Id: 9/16

FOUR 50:1 VOLTAGE CHANNELS / COMPLETE DIVIDER TO BUFFER



BUS OVERVOLTAGE REFERENCE / EXTERNAL NTC



Raw phase sense is measured before an external LC filter. Bus RC ~0.98 us; phase RC ~9.8 us.
External NTC nominal 10k; beta, physical placement and cable protection remain to be selected.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB

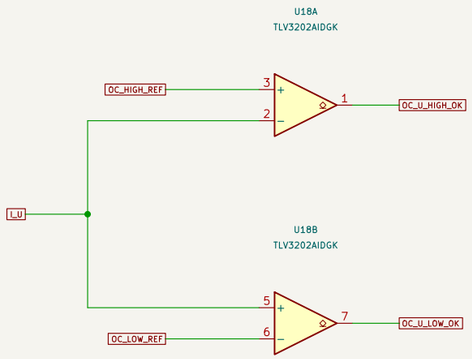
power_system

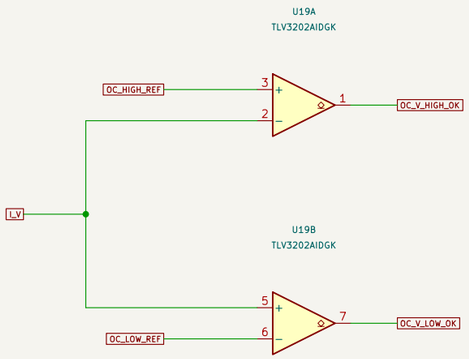
Sheet: /Voltage and NTC/
File: voltage_sense.kicad_sch

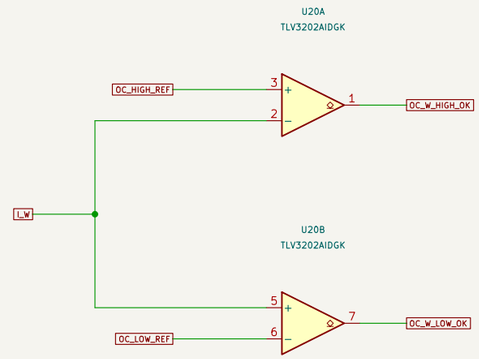
Title: Inverter - voltage and NTC sensing

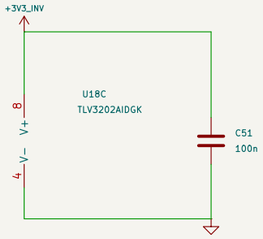
Size: A3	Date: 2026-11-06	**Rev: D-split-draft**
KiCad E.D.A. 10.0.7		Id: 10/16

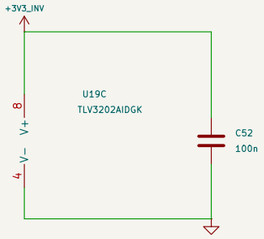
HARDWARE WINDOWS / EACH OUTPUT IS HEALTHY HIGH

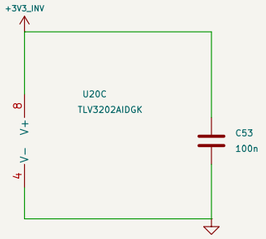




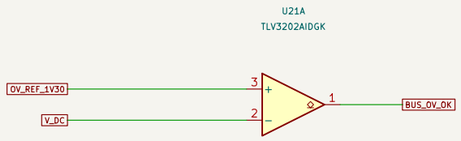


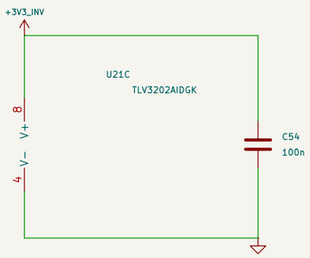


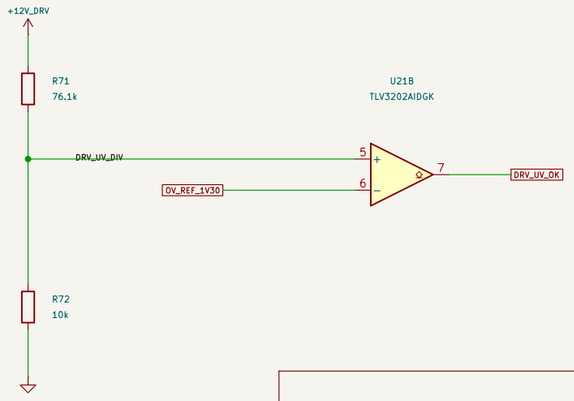




BUS OVERVOLTAGE / DRIVER RAIL UNDERVOLTAGE







Healthy = HIGH. Window violation, bus >~65 V or driver rail <~11.193 V drives its own output LOW.
Push-pull outputs enter separate AND inputs. No wired-AND; no intentional hysteresis; first event is latched.

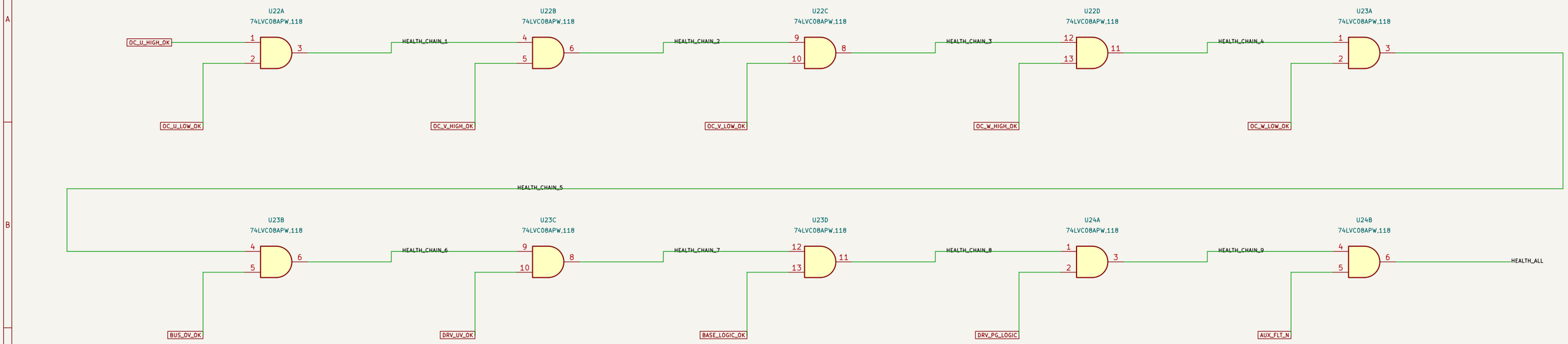
INVERTER | non-isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system

Sheet: /Analog protection/
File: comparators.kicad_sch

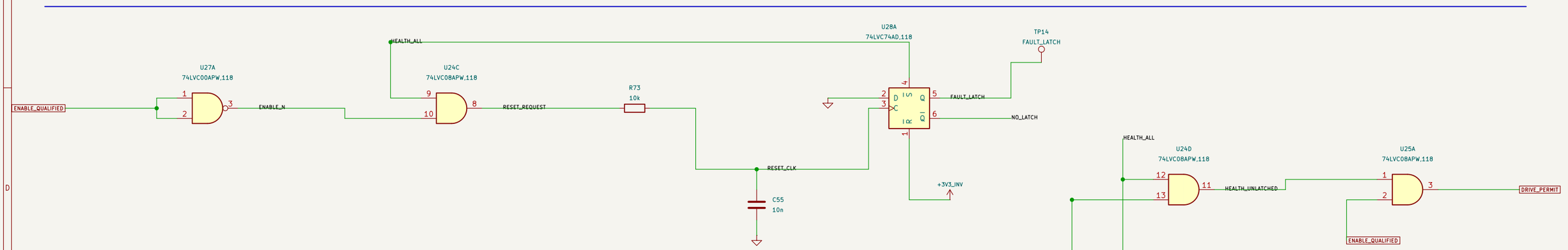
Title: Inverter — current, bus voltage and driver rail comparators

Size: A3	Date: 2026-11-06	**Rev: D-split-draft**
KiCad E.D.A. 10.0.7		Id: 11/16

HARDWARE HEALTH CHAIN / EVERY INPUT MUST REMAIN HIGH



FAULT MEMORY / DELIBERATE DISABLE–TO–RESET / PERMISSION



/S=HEALTH_ALL; /R held HIGH; D=0. Fault asynchronously sets Q.
RESET_REQUEST = HEALTH_ALL AND NOT ENABLE_QUALIFIED.
10k / 10n delays the rising reset clock: hold ENABLE low >=1 ms after recovery.
Fault or invalid supplies block DRIVE_PERMIT directly.
Recovery with ENABLE high cannot clear the latch or restart PWM.
FAULT_N is open collector; core–domain pull–up is on the MCU sub–board.
BASE_READY must be checked separately; an unpowered base cannot assert FAULT_N LOW.
Very short faults during reset may require a fresh disable cycle.

INVERTER | non–isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT – NOT RELEASED FOR PCB

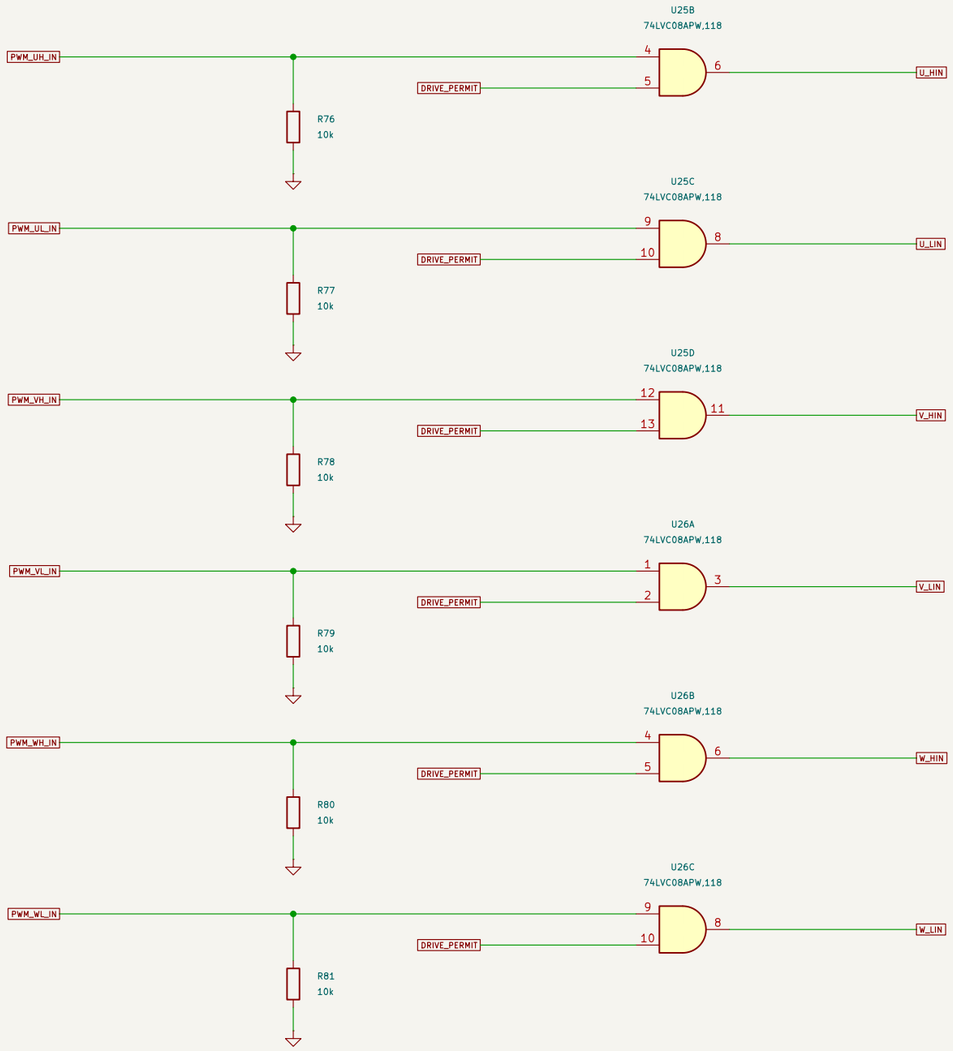
power_system

Sheet: /Hardware latch/
File: fault_logic.kicad_sch

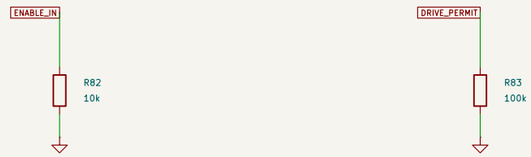
Title: Inverter – fault aggregation and reset latch

Size: A3	Date: 2026–11–06	**Rev: D–split–draft**
KiCad E.D.A. 10.0.7		Id: 12/16

HARDWARE GATING / SIX PWM COMMANDS

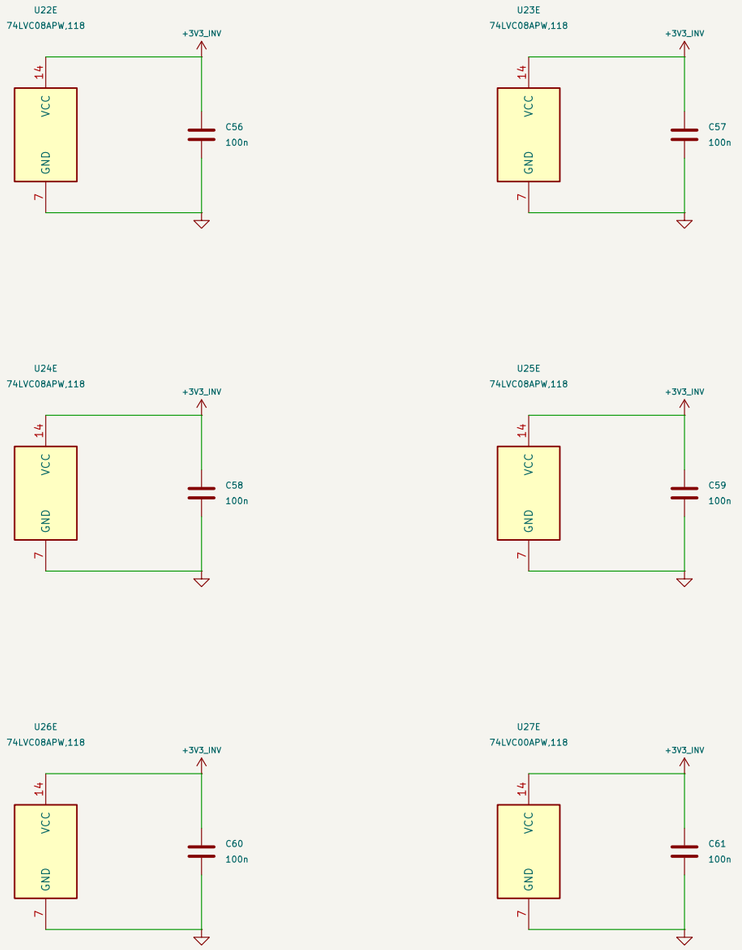


DEFAULT DISABLE / NO PERMISSION



R76–R81: hold all six commands LOW while the controller is disconnected or Hi-Z.
R82: default disabled. R83: default no permission.
Logic levels remain 3.3 V; hardware AND gates operate independently of firmware.

LOGIC POWER / LOCAL DECOUPLING



C56–C61: 100 nF local bypass at each logic package.

U26D / U27C / U27D are used on the Base interlock page.
R76–R81 default PWM LOW; R82 raw ENABLE default LOW.
No MCU on this board; CTRL_READY must qualify ENABLE before any latch reset or DRIVE_PERMIT.

INVERTER | non-isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT – NOT RELEASED FOR PCB
power_system

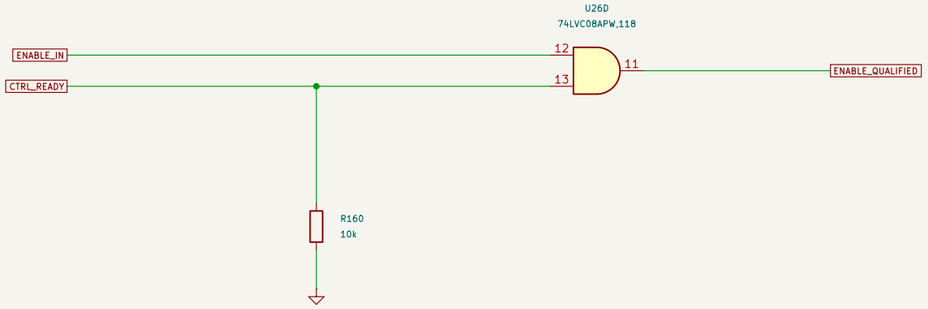
Sheet: /Hardware PWM gates/
File: control_interface.kicad_sch

Title: Inverter – local six PWM hardware gates

Size: A3	Date: 2026-11-06	**Rev: D-split-draft**
KiCad E.D.A. 10.0.7		Id: 13/16

REMOVABLE CONTROLLER / DEFAULT DISABLE

Common GND, 3.3 V logic. CTRL_READY comes from the controller reset-health buffer.
The power base owns all PWM pull-downs and hardware permission; no MCU firmware is required to block a missing controller.



R160 = 10k. Unplugged, controller reset or lost controller power => ENABLE_QUALIFIED LOW.
Only ENABLE_QUALIFIED enters the latch reset and DRIVE_PERMIT logic. Raw ENABLE_IN keeps its original local R82 pull-down.

BASE SUPPLY PRESENCE / INDEPENDENT OF OPEN-COLLECTOR FAULT



BASE_READY = BASE_LOGIC_OK AND DRV_PG_LOGIC. The MCU sub-board provides its receiver pull-down.
FAULT_N remains Q8 open collector with a pull-up only in the MCU 3.3 V domain. A high FAULT_N alone does not prove base power.
U26/U27 package power and 100 nF bypass stay on Hardware PWM gates. No added logic package.

Existing Rev C PCB is historical and does not match Rev D
POWER BASE | COMMON GND | SCHEMATIC REVIEW DRAFT
power_system

Sheet: /Base interlock/
File: base_interlock.kicad_sch

Title: Power base — controller readiness and local supply interlock

Size: A3	Date: 2026-11-06	**Rev: D-split-draft**
KiCad E.D.A. 10.0.7		Id: 14/16

LOCAL POWER INDICATOR / NO MCU LED SIGNALS ACROSS HEADER



D3: base 3.3 V present. This lamp alone does not indicate driver power or protection permission.
D4..D7 and R131..R134 belong to the removable MCU sub-board: RUN / FAULT / COMM / AUX.
The power base has no MCU-run or communication LED nets on its board-to-board header.

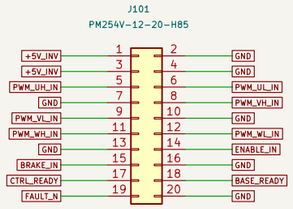
Existing Rev C PCB is historical and does not match Rev D
POWER BASE | COMMON GND | SCHEMATIC REVIEW DRAFT
power_system

Sheet: /Base LEDs/
File: base_leds.kicad_sch

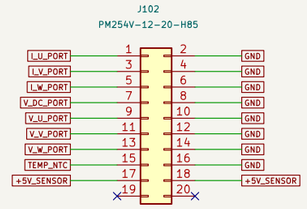
Title: Power base – local power indication

Size: A4	Date: 2026-11-06	**Rev: D-split-draft**
KiCad E.D.A. 10.0.7		Id: 15/16

J101 / DIGITAL + CORE 5V



J102 / ANALOG + SENSOR 5V



Mate J101 pin-to-pin and J102 pin-to-pin; two separate connectors. No 3V3 rail or raw NRST crosses the interface.

LOCAL SHUTDOWN REMAINS ON THIS BOARD

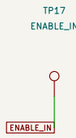
ENABLE_IN AND CTRL_READY -> ENABLE_QUALIFIED. Both inputs have local pull-downs; unplugged/reset core disables the bridge.

BASE_READY = BASE_LOGIC_OK AND DRV_PG_LOGIC. FAULT_N is the Q8 open collector; its pull-up belongs to the controller 3V3 domain.

J101 1/3 deliver +5V_INV to the core. J102 17/18 separately deliver the current-limited +5V_SENSOR branch.

J101/J102 carry no DC bus or phase current. Signal grounds return locally; keep switching-current loops away from header returns.





Mechanical review: bottom-facing male on core, top-facing socket on base; confirm pin1, rotations, standoffs and anti-misplug spacing.

Non-isolated laboratory prototype. Switch off both supplies before insertion/removal; interface clamps do not isolate a blown power stage.

NON-ISOLATED PLUG-IN SYSTEM / COMMON GND / 5 V FEED / NO HOT PLUG
HARDWARE REVIEW DRAFT – NOT RELEASED FOR PCB
power_system

Sheet: /POWER-CORE HEADERS/
File: power_interface.kicad_sch

Title: Inverter power base – mating headers

Size: A3	Date: 2026-11-06	**Rev: D-split-draft**
KiCad E.D.A. 10.0.7		Id: 16/16